

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0991

Commonwealth Chesapeake 138 kV
June 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0991. The Interconnection Customer proposes a 170.2 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on May 10, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 260 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,825,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Agawam to Quince Orchard 138.0 kV line
Contingency: Commonwealth Chesapeake to Southtown 345.0 kV
Loading: 140.6% (Rating: 171 MVA)
DFAX: 9.6%
Pre-project loading: 131.0%
2. Facility: Commonwealth Chesapeake to Rodessa 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV
Loading: 137.4% (Rating: 232 MVA)
DFAX: 12.4%
Pre-project loading: 128.4%
3. Facility: Southtown 345/138kV Transformer
Contingency: Southtown to Vienna Operations 345.0 kV
Loading: 132.0% (Rating: 1,078 MVA)
DFAX: 38.2%
Pre-project loading: 125.9%
4. Facility: Quince Orchard to Southtown 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV L
Loading: 130.4% (Rating: 336 MVA)
DFAX: 19.3%
Pre-project loading: 120.6%
5. Facility: W -Station to Shingle Crk 138.0 kV L
Contingency: Vienna Operations to Southtown 345.0 kV
Loading: 130.0% (Rating: 906 MVA)
DFAX: 53.7%
Pre-project loading: 120.0%

6. Facility: Commonwealth Chesapeake Substation to Southtown Substation 345.0 kV L
Contingency: Otay Lake to Huntington Falls 138.0 kV Line
Loading: 129.3% (Rating: 1,871 MVA)
DFAX: 72.8%
Pre-project loading: 122.7%
7. Facility: Southtown to Shingle Creek 138.0 kV
Contingency: Southtown to Vienna Operations 345.0 kV
Loading: 128.5% (Rating: 940 MVA)
DFAX: 53.7%
Pre-project loading: 118.8%
8. Facility: Sunset to Saddlebrooke Ranch 138.0 kV
Contingency: Vienna Operations to Easton 2 345.0 kV Line
Loading: 127.4% (Rating: 925 MVA)
DFAX: 42.4%
Pre-project loading: 119.6%
9. Facility: Vienna Operations Substation 345/1kV Transformer Bank
Contingency: Viena Operations to Easton 2 345.0 kV
Loading: 127.4% (Rating: 981 MVA)
DFAX: 47.6%
Pre-project loading: 119.1%
10. Facility: Quince Orchard to Colchester 138.0 kV Line
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV line
Loading: 126.9% (Rating: 556 MVA)
DFAX: 28.9%
Pre-project loading: 118.1%
11. Facility: Valleyview to Colchester 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV
Loading: 124.9% (Rating: 587 MVA)
DFAX: 28.9%
Pre-project loading: 116.5%
12. Facility: Ripple 345-138 kV Transformer
Contingency: Vienna Operations to Southtown 345.0 kV
Loading: 124.3% (Rating: 620 MVA)
DFAX: 35.9%
Pre-project loading: 114.5%
13. Facility: Prunedale to Dane 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV L
Loading: 123.7% (Rating: 303 MVA)
DFAX: 14.3%
Pre-project loading: 115.7%
14. Facility: Huntington Falls to Commonwealth Chesapeake 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV
Loading: 123.7% (Rating: 748 MVA)
DFAX: 34.6%
Pre-project loading: 115.8%
15. Facility: Oay Lake to Vedder 138.0 kV
Contingency: Commonwealth Chesapeake to Southtown 345.0 kV line
Loading: 122.8% (Rating: 1,040 MVA)
DFAX: 46.9%
Pre-project loading: 115.1%
16. Facility: Huntington Falls to Otay Lake 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV line

- Loading: 122.5% (Rating: 1,048 MVA)
DFAX: 46.9%
Pre-project loading: 114.9%
17. Facility: Tulucay to Michoud Sta 138.0 kV
Contingency: Commonwealth Chesapeake to Southtown 345.0 kV L
Loading: 122.1% (Rating: 167 MVA)
DFAX: 8.4%
Pre-project loading: 113.6%
18. Facility: Vienna Operations to Easton 2 345.0 kV
Contingency: Ripple to Linde 138.0 kV Line
Loading: 120.8% (Rating: 1,710 MVA)
DFAX: 76.9%
Pre-project loading: 113.1%
19. Facility: Langlois to Southtown 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV
Loading: 120.6% (Rating: 520 MVA)
DFAX: 27.4%
Pre-project loading: 111.7%
20. Facility: Dane to Commonwealth Chesapeake 138.0 kV
Contingency: Commonwealth Chesapeake to Southtown 345.0 kV
Loading: 119.6% (Rating: 664 MVA)
DFAX: 29.0%
Pre-project loading: 112.2%
21. Facility: Vedder to Langlois 138.0 kV
Contingency: Southtown to Commonwealth Chesapeake Substation 345.0 kV
Loading: 118.9% (Rating: 548 MVA)
DFAX: 27.4%
Pre-project loading: 110.4%
22. Facility: Memphis Junction to Persimmon Gap 138.0 kV
Contingency: Vienna Operations to Easton 2 345.0 kV line
Loading: 118.5% (Rating: 409 MVA)
DFAX: 20.2%
Pre-project loading: 110.1%
23. Facility: Grasonville to Wye Mills 138.0 kV
Contingency: Cady to Easton 2 345.0 kV
Loading: 118.3% (Rating: 203 MVA)
DFAX: 6.4%
Pre-project loading: 112.9%
24. Facility: Sunset to St Bonifacius 138.0 kV L
Contingency: Vienna Operations to Southtown Substation 345.0 kV L
Loading: 117.0% (Rating: 1,471 MVA)
DFAX: 53.7%
Pre-project loading: 110.8%
25. Facility: Tasley 345/138 kV TX
Contingency: Southtown to Commonwealth Chesapeake 345.0 kV
Loading: 116.8% (Rating: 585 MVA)
DFAX: 24.0%
Pre-project loading: 109.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Agawam to Quince Orchard 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,577,000	6% (\$164,000)
Commonwealth Chesapeake to Rodessa 138.0 kV	Rebuild line with higher-capacity conductor	\$6,090,000	6% (\$358,000)
Southtown 345/138kV TX	Replace transformer with higher-rated unit	\$12,941,000	4% (\$538,000)
Quince Orchard to Southtown 138.0 kV	Rebuild line with higher-capacity conductor	\$3,756,000	6% (\$232,000)
West -Station to Shingle Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$7,796,000	5% (\$412,000)
Southtown to Commonwealth Chesapeake 345.0 kV	Rebuild line with higher-capacity conductor	\$60,522,000	5% (\$2,991,000)
Southtown to Shingle Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$4,847,000	5% (\$256,000)
Sunset to Saddlebrooke Ranch 138.0 kV	Rebuild line with higher-capacity conductor	\$9,356,000	5% (\$443,000)
Vienna Operations Substation 345/1kV Transformer Bank	Replace transformer with higher-rated unit	\$21,104,000	5% (\$1,109,000)
Quince Orchard to Colchester 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,311,000	6% (\$143,000)
Valleyview to Colchester 138.0 kV	Rebuild line with higher-capacity conductor	\$13,610,000	6% (\$842,000)
Ripple 345-138 kV Transformer	Replace transformer with higher-rated unit	\$24,665,000	5% (\$1,321,000)
Prunedale to Dane 138.0 kV	Rebuild line with higher-capacity conductor	\$2,547,000	6% (\$158,000)
Huntington Falls to Commonwealth Chesapeake 138.0 kV	Rebuild line with higher-capacity conductor	\$2,226,000	6% (\$139,000)
Oay Lake to Vedder 138.0 kV	Rebuild line with higher-capacity conductor	\$4,009,000	6% (\$246,000)
Huntington Falls to Otay Lake 138.0 kV	Rebuild line with higher-capacity conductor	\$5,438,000	6% (\$334,000)
Tulucay to Michoud Sta Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$17,879,000	6% (\$1,137,000)
Easton 2 to Vienna Operations 345.0 kV	Rebuild line with higher-capacity conductor	\$73,558,000	5% (\$3,633,000)
Langlois to Southtown 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,900,000	6% (\$337,000)
Dane to Commonwealth Chesapeake 138.0 kV	Rebuild line with higher-capacity conductor	\$15,860,000	6% (\$981,000)
Langlois to Vedder 138.0 kV L	Rebuild line with higher-capacity conductor	\$20,343,000	6% (\$1,161,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Memphis Junction to Persimmon Gap 138.0 kV	Rebuild line with higher-capacity conductor	\$5,371,000	5% (\$259,000)
Grasonville to Wye Mills 138.0 kV	Rebuild line with higher-capacity conductor	\$16,799,000	4% (\$724,000)
Sunset to St Bonifacius 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,439,000	5% (\$250,000)
Tasley 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$12,423,000	5% (\$657,000)
	Total allocated to project		\$18,825,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.